

Pediatric Patient Counting with Interactive Object Detection

A Mathematical Framework for Occlusion-Resistant Computer Vision

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Outline of Presentation

- 1 Background & Problem
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The “Invisible Child” Phenomenon

- **Systemic Blindness:** Electronic health records (e.g., LHIMS) at KATH only capture the primary patient.
- **Cultural Variable:** Infants in Ghana are typically carried on their mothers' backs in crowded Outpatient Departments (OPDs).
- **The Data Deficit:** This undocumented influx skews hospital statistics, causing dangerous delays in triage (wait times of 2–4 hours).
- **The Bottleneck:** Standard Computer Vision fails due to the **Occlusion Problem**—overlapping spatial coordinates cause the algorithm to discard the infant as visual noise.

Foundation: Partial Occlusion in Pedestrian Detection

- **Ouyang & Wang (2015) / Wang et al.:** Established that different convolutional layers / possess distinct spatial properties for feature maps $F_l \in \mathbb{R}^{H \times W \times C}$.
- **Semantic vs. Discriminative:** Top layers maximize semantic category probability $P(obj|F_{top})$, while lower layers retain fine-grained spatial discrimination for tracking.
- **Our Application:** We utilize this hierarchical logic to isolate independent features. Instead of searching for "one person" bounding box $B \in \mathbb{R}^4$, the framework detects H and B independently to solve the cultural occlusion variable.

Foundation: FairMOT (Zhang et al., 2021)

- **The Anchor Flaw:** Previous one-shot trackers treated Re-ID as a secondary task, causing "anchor ambiguity" where multiple identities shared one bounding box.
- **The Solution:** FairMOT treats detection and Re-ID equally, utilizing an anchor-free architecture (CenterNet) to extract an embedding vector $E_{x,y} \in \mathbb{R}^D$ directly at the object's predicted center (x_c, y_c) .
- **Our Application:** By abandoning anchors and extracting features at the geometric center of the detected head, we drastically reduce Identity Switches (IDs) in the crowded OPD environment.

Foundation: PF-Track (Pang et al., 2023)

- **Spatio-Temporal Coherence:** Moving beyond frame-by-frame detection using "Past and Future Reasoning" on trajectory state vectors $V_t = [x_t, y_t, w_t, h_t]^T$.
- **Track Extension Mechanism:** Uses long-term trajectory predictions to estimate $\hat{V}_{t+\Delta t}$, maintaining the positional states of occluded objects.
- **Our Application:** If a carried infant is temporarily hidden, our system relies on predicting the Euclidean kinematics $d = \sqrt{(\hat{x}_{t+1} - x_{t+1})^2 + (\hat{y}_{t+1} - y_{t+1})^2}$ to cache their geometric footprint, preventing double-counting.

Foundation: Biological Proportions (Mlinarić et al., 2024)

- **The Canon of Proportions:** Research confirms that scale-invariant body proportions can classify age groups regardless of absolute height.
- **The Mathematical Constant:** The Dimensionless Head-to-Body Ratio:

$$R = \frac{H_{total}}{H_{head}}$$

- **Adults:** bounded strictly at $R \in [7.5, 8.0]$
- **Pediatric / Infants:** bounded at $R \in [4.0, 6.0]$
- **Our Application:** This mathematical constant serves as our primary algorithmic override, preventing the system from falsely classifying short adults as pediatric patients.

Synthesis: The "One Body, Two Identities" Logic

By merging these four pillars, our proposed framework operates as follows:

- 1 **Detect:** Isolate independent Head (H) and Body (B) bounding boxes.
- 2 **Trigger:** If one Body shape bounds two distinct Heads ($h_1, h_2 \in H$), trigger spatial evaluation.
- 3 **Evaluate Spatial Displacement:** Verify $y_{h2} < y_{h1}$ (Vertical Displacement) and $A_{h2} < A_{h1}$ (Area Proportion).
- 4 **Verify Proportion:** Apply the Allometric Ratio $R = \frac{H_{total}}{H_{head}}$. If $R < 6.0$, confirm Pediatric status.

Edge Computing & Data Ethics

- **Volatile Processing:** All bounding box inferences and kinematic tracking ($d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$) occur entirely in local RAM.
- **No Image Storage:** Video frames are immediately destroyed after spatial coordinates are extracted.
- **JSON Output:** The only data transmitted to the hospital dashboard is an anonymized numerical matrix (e.g., {"adults": 12, "occluded_infants": 2}).

Conclusion

- We are bridging the gap between hospital records and physical reality in Ghanaian healthcare.
- By combining **Anthropometric Constraints** (Mlinarić), **Anchor-free Detection** (FairMOT), and **Spatio-Temporal Kinematics** (PF-Track), we correct standard computer vision occlusion failures.
- We transform the "Invisible Child" into actionable, privacy-compliant data to optimize hospital triage and significantly reduce patient wait times.

Thank You.

Questions?